

Rate My Professor (RMP) Data Analysis: Gender Bias and Predictive Modeling

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0 Exploratory Data Analysis (EDA) and Pre-Processing

The pre-processing and EDA generalize to all questions. However, for some questions, there may be extra pre-processing. An overview of the pre-processing involves, first, merging the three given datasets (tags, ratings, and qualitative). This first step also includes handling missing values to ensure data integrity. For example, dropping missing values before merging the ratings dataset could lead to misalignment, where rows in the qualitative (or tags) dataset no longer correspond to those in the ratings dataset. The second step involves the normalization of the tags variables. A professor with more ratings will have received more tags. Therefore, we have normalized the tags. Lastly, variables with averages, such as rating and difficulty, may have extreme values if a professor only has a few ratings. The following subsections detail the pre-processing steps.

0.1 Merging, Missing Values, and Inconsistencies

To ensure reproducibility and consistency in the results of stochastic processes such as data splitting, sampling, and model initialization, a random seed value of 12299542 was set at the beginning of the analysis.

Since all the datasets have the same number of rows (89893), we concatenated the three dataframes which resulted in a dataset of shape (89893, 31). Only after merging do we handle missing values in order to preserve the integrity of the relationships across datasets.

None of the tags variables contain missing values, and across all 31 variables, only nine variables contain missing values. Across all of those nine variables, they shared 19889 NaN; therefore, we dropped the rows associated with those NaN values. One variable, **Take Again Proportion**, had a total of 70,000 NaN, however, we only dropped the 19889 since if had dropped all the NaN values, we would lose a big portion of the data.

After merging and addressing most missing values, we resolved inconsistencies within the merged dataframe. First, the **US State** variable in the qualitative data was expected to contain only two-letter state abbreviations. However, 133 entries were in different formats, representing non-US cities such as London, Manchester, and Glasgow. These rows were removed.

Second, we addressed inconsistencies in the **Male** and **Female** binary variables. While 15,664 professors were neither male nor female—assumed to be non-binary rather than erroneous—2,207 professors were marked as both male and female simultaneously, which is logically inconsistent. These rows were removed as the two categories are mutually exclusive.

After handling missing values and inconsistencies, the dataset was reduced to 67,664 rows.

0.2 Normalization

The tags dataset is based on raw counts, meaning professors with more ratings are likely to have more tags. To account for this bias, we normalized the tags by dividing the number of tags by the total number of ratings for each professor as shown in Equation 1.

$$\text{Normalized Tag}_{ij} = \frac{\text{Tag Count}_{ij}}{\text{Total Ratings}_i} \quad (1)$$

where Tag Count_{ij} represents the count for tag j for professor i , and Total Ratings_i is the total number of ratings for professor i .

0.3 Bayesian Average: Handling Averages based on Few Ratings

When working with average ratings (or difficulty), it is important to account for the bias introduced by the number of ratings a professor has received. A professor with very few ratings may have an artificially high or low average score due to the limited sample size. For example, a professor with a single 5-star review would rank at

the top, while a professor with a single 1-star review would rank at the bottom. These extremes do not provide meaningful insights due to insufficient data.

To address this, instead of using a minimum threshold for the number of ratings, we use the Bayesian Average Rating, which incorporates a prior belief and adjusts the average rating based on the number of ratings received. The formula is given by:

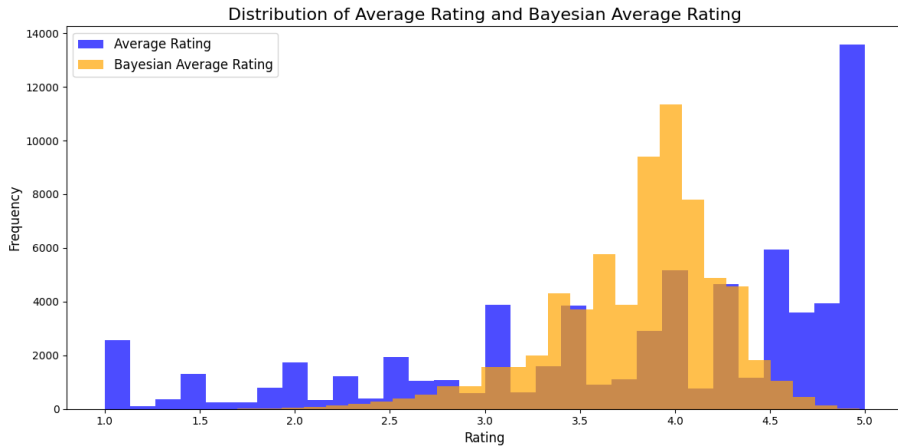
$$\text{Bayesian Average Rating} = \frac{\mu \cdot v + n \cdot \bar{r}}{v + n}$$

where:

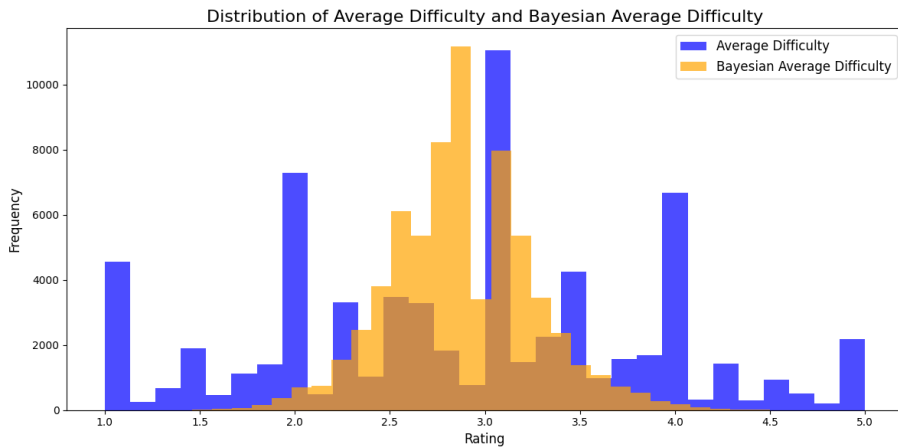
- $\mu = 2.867$ is the prior mean (overall average rating across all professors, calculated from the dataset).
- v is the virtual count, representing the strength of the prior (the weight given to the prior mean).
- n is the number of ratings for the professor.
- $\bar{r}$ is the professor's average rating based on their received ratings.

This approach prevents professors with very few ratings from being ranked at extreme positions. For example, a professor with no reviews starts with an average rating of $\mu = 2.867$, which reflects the prior belief. Each new rating slightly adjusts the average. For instance, a 5-star review will increase the Bayesian average, while a 1-star review will decrease it, but the effect diminishes as the number of reviews grows.

By using this method, the Bayesian average balances the prior knowledge of overall average ratings with individual data, ensuring that professors with a small number of ratings do not dominate the rankings disproportionately. We do this for both average ratings and difficulty. Figure 1 compares the distributions of the averages with Bayesian averages.



(a) Professors' Average Ratings



(b) Professors' Average Difficulty

Figure 1: Comparison of Bayesian metrics for professors' ratings

Observe from the panel above that the Bayesian adjustment smooths the distribution of difficulty ratings by reducing extreme fluctuations, pulling values toward the overall mean, and providing a more reliable metric for comparison, especially for professors with fewer ratings.

1 Gender Bias in Student Evaluation of Professors

To answer the question of whether there is evidence of a pro-male gender bias in this dataset we will use a significance test.

We first extracted the female and male professor groups along with their average ratings. Given that the average ratings don't follow a normal distribution, as can be seen in Figure 2, we opted for a non-parametric approach. Specifically, we used the Mann-Whitney U test since it does not assume that the two groups distribute normally or have equal variances. Moreover, since we are interested in whether there is evidence of a pro-male gender bias in this dataset, we used a one-tailed test. We then formulated our hypotheses as follows:

- **Null Hypothesis (H_0):** There is no significant difference between female and male professors average ratings.
- **Alternative Hypothesis (H_a):** Male professors average ratings are significantly higher than female professors average ratings.

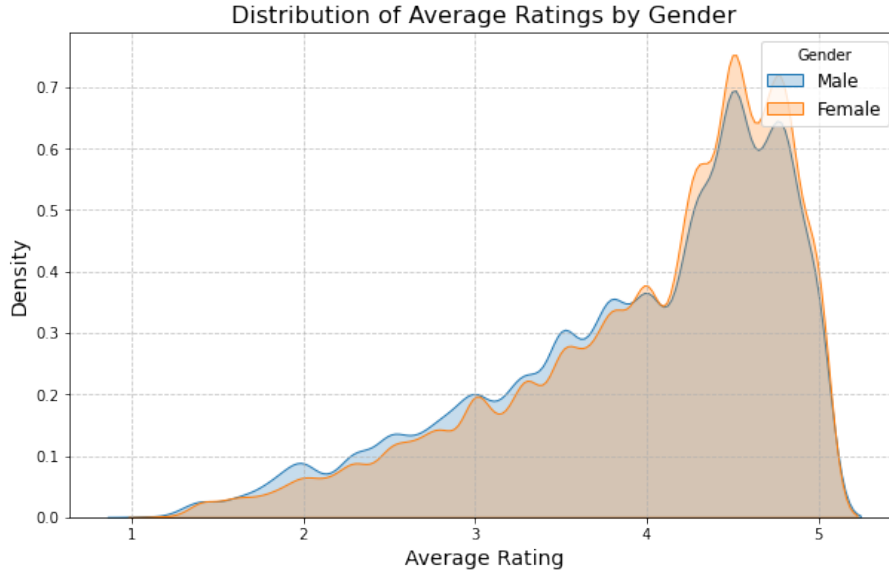


Figure 2: Average Ratings for Males and Females

The results of the Mann-Whitney U test revealed a statistically significant difference between the groups, with a p-value of $p = 4.22 \times 10^{-5}$ which is smaller than the significance level, $\alpha.005$. Therefore, we reject the null hypothesis and conclude that there is evidence of a pro-male gender bias in this dataset.

However, to ensure that the average rating is solely dependent on gender we looked for possible confounders by running a two steps approach:

1. **Step 1: Testing Features by Gender:** split all features except for strings and average rating by gender and using Mann-Whitney U test to identify features that are pro-male and store them for step two.
2. **Step 2: Testing Features' Impact on Average Rating:** For each feature from step 1, we identified a threshold which is the median of the feature to divide the data into two groups based on whether they are above or below the threshold. Then we used a one-sided Mann-Whitney U test to check if Average Rating was significantly higher in the high group compared to the low group. For example, if Hilarious has higher values for male professors, we wanted to test whether Hilarious contributed to higher ratings.

After running these two steps we found that many features differ significantly by gender. Moreover, these same features also significantly affect the Average Rating. Since other features like Respected and Hilarious mediate the relationship between gender and Average Rating, we conclude that there isn't enough evidence of a pro-male gender bias in this dataset.

2 Gender Difference in the Spread of the Ratings Distribution

To assess the equality of variances between the rating distributions of female and male professors we used Levene test since it does not assume normality of the data and we formulated our hypotheses as follows:

- **Null Hypothesis (H_0):** There is no significant difference between female and male professors average rating distributions.
- **Alternative Hypothesis (H_a):** Male professors' average rating variance significantly differs from female professors' average rating variance.

The results of Levene test revealed a statistically significant difference between the variances of the groups, with a p-value of $p = 0.0003$ which is smaller than the significance level, $\alpha.005$. Therefore, we reject the null hypothesis and conclude that the variances of male and female average ratings are significantly different.

Since we didn't cover the Levene test in this course, we thought it would be interesting to apply the tools we learned throughout the semester in a less straightforward way, so we applied a permutation test to test gender difference in the spread of the ratings distribution with the same hypotheses:

- **Null Hypothesis (H_0):** There is no significant difference between female and male professors average rating distributions.
- **Alternative Hypothesis (H_a):** Male professors' average rating variance significantly differs from female professors' average rating variance.

To assess whether there is a significant difference in the variances of ratings between genders using permutation test we began by combining the ratings of female and male professors into a single array and defined the number of observations for each group. The test statistic chosen was the difference in variances between the two groups. To estimate the null distribution, we repeatedly permuted the combined ratings, splitting them into two groups of the same sizes as the original datasets, and computed the test statistic for each permutation and the process 10,000 times.

To determine whether there is a significant difference between the two groups, we need to account for deviations on both ends of the null distribution. This is done by taking the absolute values of the test statistics, ensuring that differences in both directions are considered. To calculate the p-value, we compared the observed test statistic (calculated from the original data) to the test statistics generated from the permutations, which form the null distribution. Afterward, we calculated the proportion of permuted test statistics whose absolute values are greater than or equal to the absolute value of the observed test statistic.

Surprisingly, unlike Levene's test using the permutation test, we got a p-value $p = 0.0063$, so we fail to reject the null hypothesis and conclude that the variances of male and female average ratings are not significantly different. Moreover, we suspect that these differences could be due to how we chose to define our permutation test.

3 Effect Sizes of gender bias in average rating and gender bias in spread of average rating

To estimate the effect size of these effects we used different variations of Cohen's d and quantified uncertainty through bootstrapping to obtain a 95% confidence interval.

3.1 Effect Size of Gender Bias in Average Rating

Cohen's d

The formula for Cohen's d is given by:

$$d = \frac{\mu_1 - \mu_2}{\text{pooled_std}}$$

where the pooled standard deviation (pooled_std) is:

$$\text{pooled_std} = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

Here:

- μ_1, μ_2 : Mean of female and male professors average ratings.
- s_1, s_2 : Standard deviation of female and male professors average ratings.
- n_1, n_2 : Sample size of female and male professors average ratings.

Bootstrapping for Cohen's d

We then followed the following approach to bootstrap:

1. Draw bootstrap samples with replacement from female and male professors average ratings
2. Compute Cohen's d for each bootstrap sample:
3. Collect all bootstrap values to form the bootstrap distribution.
4. Calculate the 95% confidence interval by determining the range between the 2.5th percentile and the 97.5th percentile of the bootstrapped Cohen's d values.

Results

The calculated Cohen's d was 0.090, with a 95% confidence interval ranging from 0.048 to 0.131 as can be seen in Figure 3. These results suggest a small effect size suggesting that there is a slight difference in the average ratings between the two groups.

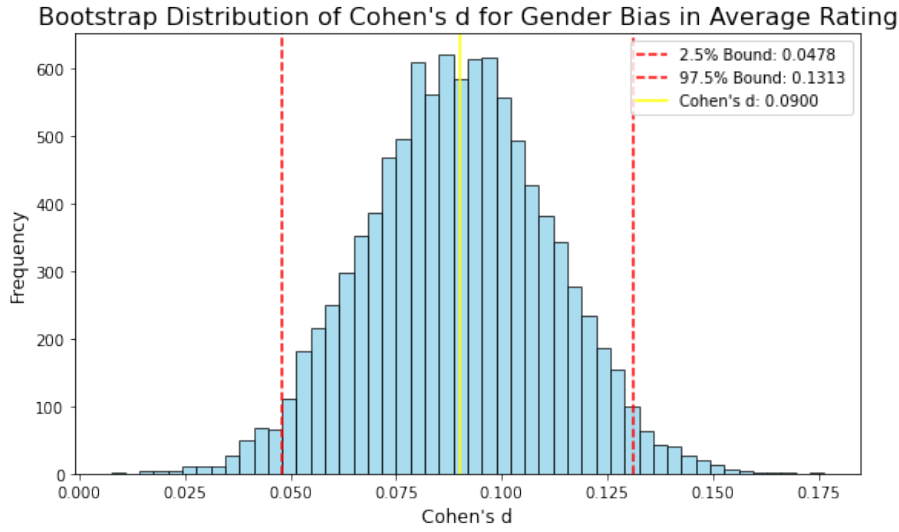


Figure 3: Effect Size of Gender Bias in Average Rating

3.2 Effect Size of Gender Bias in Spread of Average Rating

Here, we calculated the effect size for the difference in variance (spread) between male and female average ratings using a modified Cohen's d.

Cohen's d

$$\text{Effect Size} = \frac{\text{Var}_1 - \text{Var}_2}{\text{pooled_std}}$$

where the pooled standard deviation (pooled_std) is:

$$\text{pooled_std} = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

Here:

- μ_1, μ_2 : Variance of female and male professors average ratings.
- s_1, s_2 : Standard deviation of female and male professors average ratings.
- n_1, n_2 : Sample size of female and male professors average ratings.

Bootstrapping for Cohen's d

We then followed the following approach to bootstrap:

1. Draw bootstrap samples with replacement from female and male professors average ratings.
2. Compute Cohen's d for each bootstrap sample:
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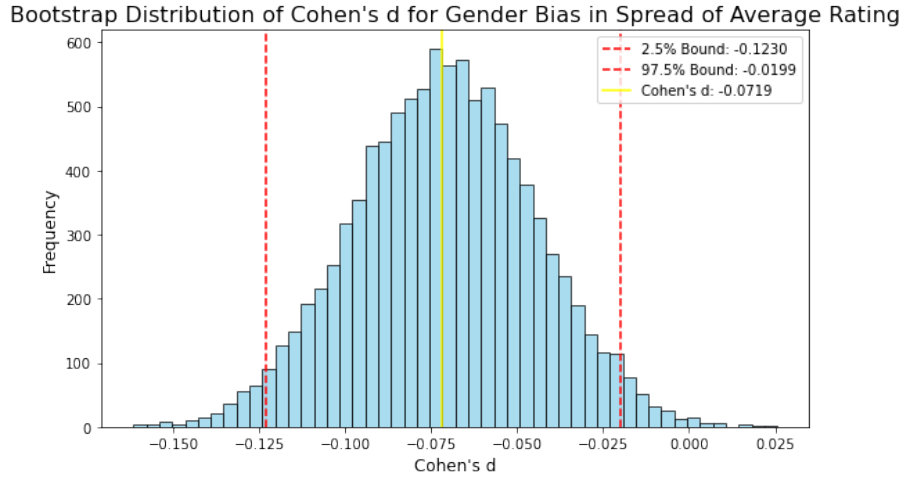


Figure 4: Effect Size of Gender Bias in Average Rating

The analysis reveals a small but statistically significant gender bias in the spread of average ratings, with an effect size of -0.0719 and a 95% confidence interval of $(-0.1230, -0.0199)$ as can be seen in Figure 4. This indicates that male ratings have slightly less variability compared to female ratings.

4 Gender Difference in the Tags Awarded by Students

- **Null Hypothesis (H_0):** There is no significant difference between male and female professors in the tags awarded by students.
- **Alternative Hypothesis (H_A):** There is a significant difference between male and female professors in the tags awarded by students.

To examine gender differences among the 20 tags awarded by the students, entries with fewer than the median number of ratings were excluded to ensure the analysis focused on professors with sufficient evaluations. The data distributions were visualized and found to be skewed, violating the assumption of normality. Consequently, the Mann-Whitney U test was chosen due to its suitability for non-normally distributed data and its ability to compare medians between two independent groups without assuming equal variances.

The Mann-Whitney U test identified statistically significant gender differences ($p < 0.005$) for 19 out of the 20 tags. The updated results are summarized in the table below, which highlights the three tags with the strongest evidence of gender differences (lowest p -values) and the three tags with the weakest evidence (highest p -values).

Table 1: Results of Mann-Whitney U Tests for Gender Differences in Tags

Tag	U-Statistic	p-Value	Significance
Hilarious	130,823,899.5	3.71×10^{-228}	Yes
Amazing Lectures	120,034,116.0	5.51×10^{-54}	Yes
Lecture Heavy	118,242,699.0	3.05×10^{-39}	Yes
Caring	100,751,939.5	1.42×10^{-37}	Yes
Respected	118,619,220.0	5.02×10^{-35}	Yes
Participation Matters	101,350,254.0	3.87×10^{-34}	Yes
Good Feedback	101,236,459.0	4.00×10^{-33}	Yes
Graded by Few Things	115,480,384.0	2.80×10^{-31}	Yes
Group Projects	104,731,292.5	1.65×10^{-24}	Yes
Lots of Homework	104,262,502.5	2.96×10^{-17}	Yes
So Many Papers	106,446,050.5	1.46×10^{-14}	Yes
Extra Credit	105,246,648.5	1.56×10^{-14}	Yes
Test Heavy	113,102,672.0	3.18×10^{-12}	Yes
Lots to Read	106,584,859.5	6.33×10^{-7}	Yes
Inspirational	113,029,601.0	1.99×10^{-6}	Yes
Accessible	112,797,869.5	2.21×10^{-6}	Yes
Clear Grading	106,922,336.5	1.27×10^{-5}	Yes
Tough Grader	106,935,847.0	1.34×10^{-5}	Yes
Don't Skip Class	107,293,280.0	1.30×10^{-4}	Yes
Pop Quizzes!	110,916,313.0	2.66×10^{-2}	No

The most significant differences were observed in subjective tags like **Hilarious**, **Amazing Lectures**, and **Lecture Heavy**, which had the lowest p-values. These findings suggest that attributes like charisma, lecture delivery, and engagement are strongly influenced by the professor's gender, likely reflecting societal stereotypes about communication and emotional intelligence. On the other hand, the tag **Pop Quizzes!**, which had the highest p-value ($p = 0.0266$), showed no significant gender differences, indicating that it may be less affected by gender-related biases. Overall, the results highlight the influence of gender on how subjective tags are assigned, while certain instructional characteristics, like the presence of pop quizzes, appear to be perceived more objectively.

5 Gender Difference in Terms of Average Difficulty?

- **Null Hypothesis (H_0):** There is no significant difference in "Bayesian Average Difficulty" ratings between male and female professors.
- **Alternative Hypothesis (H_A):** There is a significant difference in "Bayesian Average Difficulty" ratings between male and female professors.

To determine whether there is a significant difference in "Bayesian Average Difficulty" ratings between male and female professors, the median of the "Number of Ratings" was calculated, and entries with ratings below the median were excluded to ensure the analysis focused on professors with sufficient evaluations. This filtering step reduced the potential bias associated with limited data points.

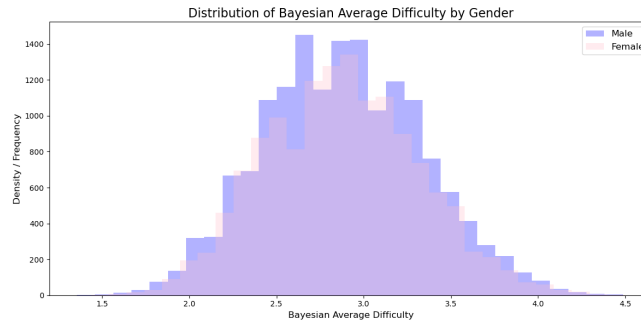


Figure 5: Distribution of Bayesian Average Difficulty by Gender

As shown in Figure 3, the distributions of the ratings appear approximately normal. Consequently, an independent samples t-test, which assumes normality and independence of the samples, was conducted.

The analysis yielded a t-statistic of 0.293 and a p-value of 0.7695, indicating no statistically significant difference in average difficulty ratings between male and female professors. These results suggest that gender does not influence how professors are rated in terms of difficulty. Therefore, the null hypothesis, which posits no difference in difficulty ratings between male and female professors, is not rejected.

6 Effect Size

- **Null Hypothesis (H_0):** There is no effect of gender on the “Bayesian Average Difficulty” ratings (*Cohen's $d = 0$*).
- **Alternative Hypothesis (H_A):** There is a nonzero effect of gender on the “Bayesian Average Difficulty” ratings (*Cohen's $d \neq 0$*).

To quantify the likely size of the effect, Cohen’s d was calculated as a measure of the standardized mean difference between male and female professors’ “Bayesian Average Difficulty” ratings. Bootstrapping was applied with 10,000 resamples to estimate a 95% confidence interval for Cohen’s d , providing a robust assessment of the effect size.

Cohen’s d was chosen to quantify the magnitude of the difference between the two groups, as it is a widely used effect size measure. Bootstrapping was used to avoid reliance on parametric assumptions, ensuring that the confidence intervals for the effect size are data-driven and robust to deviations from normality.

The calculated Cohen’s d was 0.0034, with a 95% confidence interval ranging from -0.0191 to 0.0260. These results suggest a negligible effect size, as the confidence interval includes values close to zero, indicating no meaningful difference between male and female professors’ difficulty ratings. Figure 4, which visually shows the bootstrap distribution, confirms the tight clustering of values around zero.

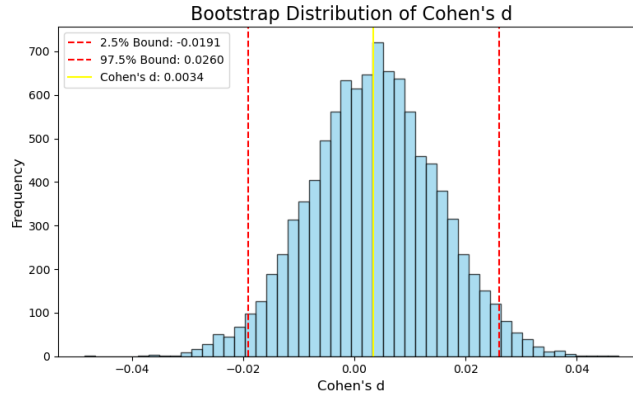


Figure 6: Bootstrap Distribution of Cohen’s d

Given that Cohen’s d is close to zero and the 95% confidence interval includes values near zero, it can be concluded that the effect size is negligible. This supports the conclusion that gender has no meaningful influence on “Bayesian Average Difficulty” ratings. Limitations include the assumption of independence and the potential influence of unmeasured variables, but the bootstrapping approach mitigates concerns about distributional assumptions.

7 Regression: Predict *Average Rating* based on Numerical Predictors

The goal of the first regression task is to predict *average rating* based on numerical predictors. Two considerations for the regression; first, after our pre-processing (in Section 0), all of our numerical predictors have zero missing values except for the **Take Again Proportion** feature which still has 55,871 missing values out of 67,664 rows. We can either drop the missing values, impute them with the mean (or median), or drop the feature. Second, based on the correlation matrix of the features in Figure 13 (in the Appendix section), we can see that there are three pairs of features with moderate correlation ($\pm 0.4 - \pm 0.5$) including Bayesian average difficulty and take again proportion. Due to those two considerations, we built five regression models and Table 2 below summarizes the metrics for each model. A 0.8/0.2 train/test split was applied.

From the non-regularized models, dropping the missing values performed the best across all metrics which implies that the **Take Again Proportion** feature is important. Imputing the many missing values based on the

Model	R^2	RMSE	Feature 1	Feature 2	Feature 3
Drop NaN	0.81	0.38	Bayesian Average Difficulty (-0.29)	Received Pepper (0.20)	Male (0.06)
Impute NaN	0.32	0.92	Bayesian Average Difficulty (-1.28)	Received Pepper (0.55)	Male (0.15)
Drop Feature	0.31	0.93	Bayesian Average Difficulty (-1.37)	Received Pepper (0.60)	Male (0.16)
LASSO $\lambda = 0.0013$	0.81	0.45	Take Again Proportion (0.735)	Bayesian Average Difficulty (-0.175)	Received Pepper (0.118)
Ridge $\lambda = 2.5$	0.81	0.45	Take Again Proportion (0.735)	Received Pepper (0.119)	Bayesian Average Difficulty (-0.176)

Table 2: Performance Metrics and Most Predictive Features for Regression Models Predicting Average Rating Using the Ratings Dataset

mean seems to have introduced bias which increased the RMSE whereas dropping the feature entirely cost us the loss of important information. The three most protective features are consistent across the non-regularized models. The regularized models, on the other hand, perform equally well to the first model (dropping NaN) which implies that the effect of multicollinearity is not extreme. This is further supported by the small penalty terms (λ) for both regularized models.

While the predictive powers of models 1,4, and 5 are similar, the most predictive coefficients differ. Non-regularized models rely more on Bayesian average difficulty, suggesting that difficulty perceptions are a primary driver of average ratings when features are assessed independently. Regularized Models shift to Take Again Proportion, highlighting its robustness as a predictor when shared variance with other features (like Bayesian average difficulty) is minimized. Given all the metrics, if the goal is precise prediction, then model 1 might seem preferable due to its lower RMSE. However, if robustness of coefficients is more important, a regularized model, specifically LASSO, is likely a better choice despite its higher RMSE. Figure 7 visualizes the outcomes and predicted outcomes for model 1.

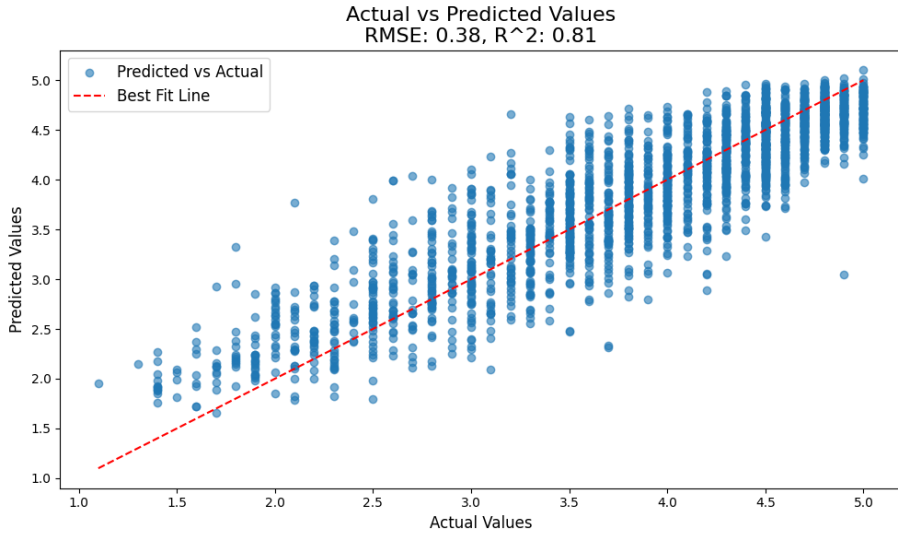


Figure 7: Scatterplot of Model 1 of Actual and predicted outcomes

8 Regression: Predict *Average Rating* based on Tags

The goal of the second regression task is to predict *average rating* from all tags in the tags dataset. We used the normalized tags as explained in the pre-processing subsection 0.2. Similar to the previous model, a train/test split of 0.8/0.2 was applied to test the model's generalizability to new data. Since we are using tags and not averages, the Bayesian averages that were constructed do not help here; therefore, to reduce noise in the data, a minimum threshold of five number of ratings was applied. Meaning we only considers professors who received five or more ratings. Figure 8a shows that the correlation between tags features is generally low, and only four pairs of features have a correlation around ± 0.3 which alleviates multicollinearity concerns. Further, Figure 8b compares the actual ratings to the predicted ratings as well as presents the R^2 and RMSE.

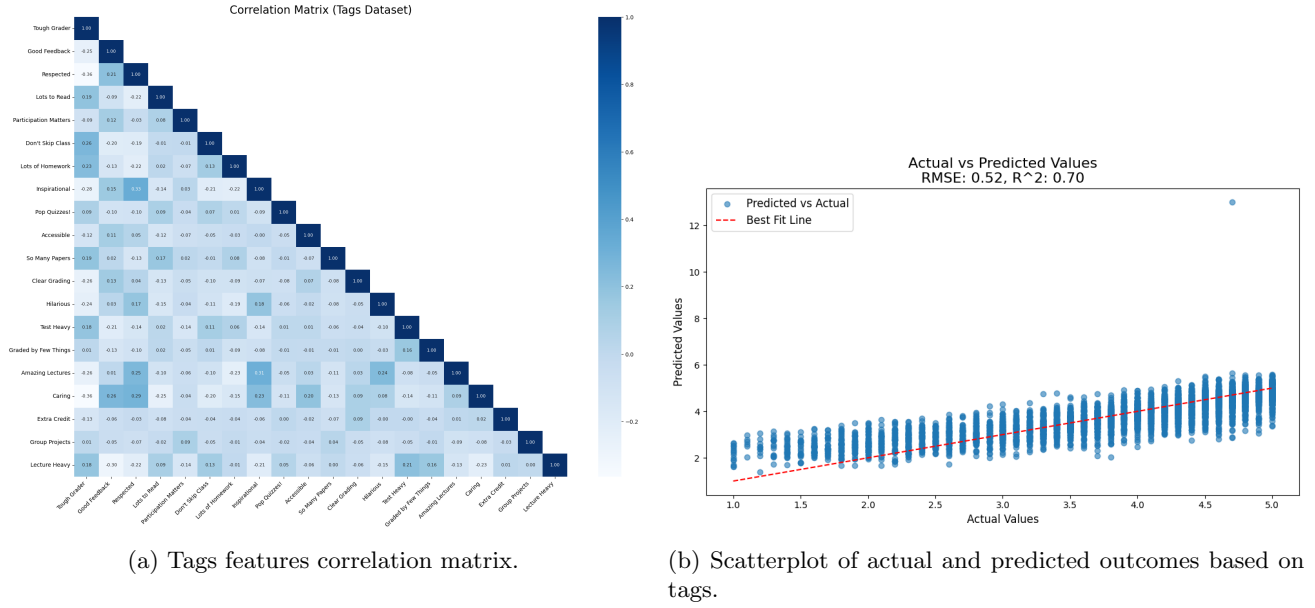


Figure 8: From the correlation matrix, pairs that had a correlation ± 0.3 include **Tough Grader** and **Respected**.

From Figure 8b, observe that the RMSE is 0.7 which suggests that 70% of the variance in average ratings is explained by the tags. The RMSE of 0.52 indicates that the model's predictions are reasonable. Despite multicollinearity not being a main concern for these given set of features, two regularized regression models were still built. Table 3 summarizes the metrics and presents the most predictive tags for average rating.

Model	R^2	RMSE	Feature 1	Feature 2	Feature 3
Linear Regression	0.7	0.52	Amazing Lectures (1.20)	Good Feedback (1.02)	Respected (1.01)
LASSO $\lambda = 0$	0.7	0.54	Tough Grader (-0.24)	Good Feedback (0.24)	Amazing Lectures (0.20)
Ridge $\lambda = 33.5$	0.7	0.54	Tough Grader (-0.24)	Good Feedback (0.24)	Amazing Lectures (0.20)

Table 3: Performance Metrics and Most Predictive Features for Regression Models Predicting Average Rating Using Tags

Based on the summary above, all three models have the same $R^2 = 0.7$, indicating that 70% of the variance in average ratings is explained by the tags. Linear Regression achieves the lowest RMSE (0.52), suggesting slightly better predictive accuracy on unseen data compared to LASSO and Ridge, which both have RMSE values of 0.54. However, the regularized models exhibit more robust coefficient stability, which is particularly beneficial given the number of features. The most predictive features are consistent among the regularized models with **Tough Grader** being the most significant (negative) predictor.

Despite the low correlation between features, regularization helps stabilize the coefficients, enhances robustness by preventing overfitting, and mitigates potential noise in the data. Despite a slightly higher RMSE in the regularized models, they are still preferred in this context for the stated reasons. The choice between LASSO and Ridge depends on the importance of feature selection. LASSO would be preferred when simplifying the model is a priority, as it shrinks some coefficients to zero.

9 Regression: Predict Average Difficulty

The goal of the third regression task is to predict *average difficulty* from all tags in the tags dataset (same features as in Section 8). Similar to the previous section, we used the normalized tags, set a minimum threshold to the number of ratings of five (to reduce noise), and applied a train/test split of 0.8/0.2 to test the model's generalizability to new data. Since we are using the same tags features as the previous question, colinearity is not a main concern as was seen in the correlation matrix in Figure 8a. However, given the many variables, we still applied regularized regression to have more stable coefficients. Table 4 below summarizes the metrics and presents the most predictive tags for average difficulty.

All three models achieved an $R^2 = 0.53$ indicating that all three models capture 53% of the variance in average difficulty. However, the RMSE is higher in the regularized models (LASSO and Ridge) at 0.68. This suggests

Model	R^2	RMSE	Feature 1	Feature 2	Feature 3
Linear Regression	0.53	0.55	Tough Grader (1.70)	Test Heavy (1.22)	Accessible (0.88)
LASSO $\lambda = 0.0001$	0.53	0.68	Tough Grader (0.49)	Accessible (0.12)	Test Heavy (0.12)
Ridge $\lambda = 37$	0.53	0.68	Tough Grader (0.49)	Accessible (0.12)	Test Heavy (0.12)

Table 4: Performance Metrics and Most Predictive Features for Regression Models Predicting Average Difficulty Using Tags

that regularization, while beneficial for stabilizing coefficients, led to a small trade-off in predictive accuracy in this context.

From Table 4, **Tough Grader** consistently emerges as the most predictive feature across all models, with a coefficient of 1.34 in Linear Regression and approximately 0.49 in both LASSO and Ridge. Notice that the top three predictive features of average difficulty are the same for the linear regression and regularized models but with slightly different order. While regularization reduces overfitting and stabilizes coefficients, it comes with a trade-off in this case, as seen in the higher RMSE. The choice between the models depends on the specific context: Linear Regression provides the best predictive accuracy for this task, whereas regularized models may be preferred if feature stability and interpretability are prioritized. Figure 9 shows the actual difficulty rating compared to the predicted average difficulty using tags for the first model (linear regression).

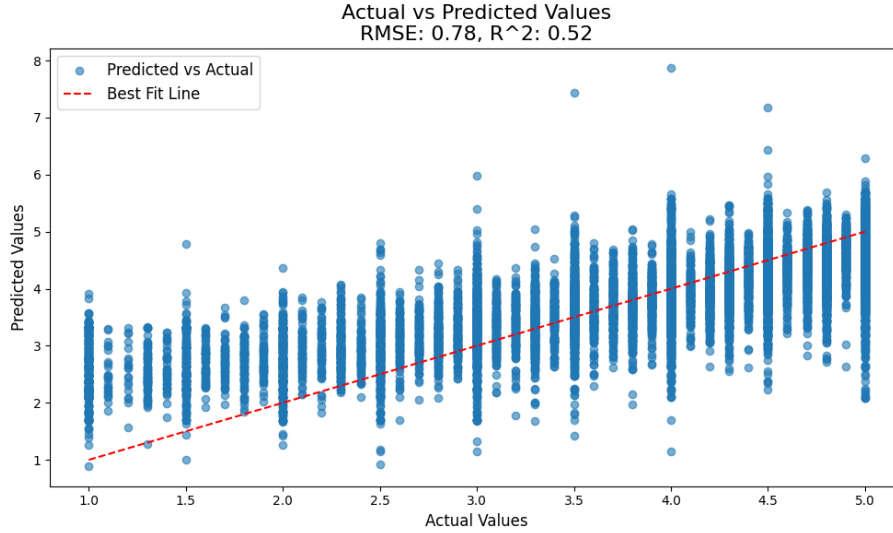


Figure 9: Scatterplot of Model 1 of Actual and predicted outcomes

10 Classification: Logistic Regression and Linear SVC

To predict whether a professor receives a "pepper," several classification models were designed and tested using different preprocessing approaches. Since the "Take Again Proportion" column had 82.57% missing data, two strategies were explored: dropping rows with missing values in this column ("Drop NaN") and removing the column entirely ("Drop Column"). Principal Component Analysis (PCA) was also applied in some models to evaluate the impact of dimensionality reduction, given the dataset's high dimensionality with 30 features. The median of the "Number of Ratings" was calculated, and entries with ratings below the median were also excluded to ensure the analysis focused on professors with sufficient evaluations.

Class imbalance, a recurring challenge in classification, was addressed using the SMOTE algorithm to balance the representation of both classes (professors with and without peppers) in the training data. This step ensured that the models were not biased toward the majority class. Metrics such as AUC, accuracy, precision, recall, and weighted F1-score were used to thoroughly evaluate model performance.

Table 5: Classification Model Performance Metrics

Model	AUC	Accuracy	Precision (0)	Recall (0)	Precision (1)	Recall (1)	F1-Score (Weighted)
Logistic Regression (Drop NaN)	0.8072	74%	0.76	0.69	0.72	0.78	0.74
Logistic Regression (Drop Column)	0.7829	71%	0.74	0.66	0.69	0.77	0.71
Linear SVC (Drop NaN)	0.8074	73%	0.76	0.68	0.71	0.79	0.73
Linear SVC (Drop Column)	0.7828	71%	0.74	0.65	0.69	0.78	0.71
Logistic Regression with PCA	0.7927	72%	0.73	0.67	0.70	0.76	0.72
Linear SVC with PCA	0.7931	72%	0.74	0.67	0.70	0.76	0.72

The analysis showed that Logistic Regression and Linear SVC performed best when the "Take Again Proportion" column was retained and rows with missing values were removed ("Drop NaN"). These models achieved AUC scores of 0.8072 and 0.8074, respectively, with accuracies of 74% and 73%. This underscores the importance of the "Take Again Proportion" column, even with its missing data, and highlights that careful handling of missing values preserves meaningful insights.

In contrast, removing the column entirely led to a slight performance decline (AUC 0.7829), reaffirming its predictive value. This result emphasizes a broader insight: even features with substantial missingness can provide significant information if handled appropriately, through methods such as imputation or selective removal.

Using PCA with 23 components to retain 95% of the variance resulted in a marginal drop in performance. Logistic Regression and Linear SVC with PCA achieved AUC scores of 0.7927 and 0.7931, respectively, with accuracies of 72%. While PCA reduced the feature set and computational complexity, these results suggest that retaining the full set of features enhances predictive accuracy, particularly when the features have strong individual contributions to the classification task.

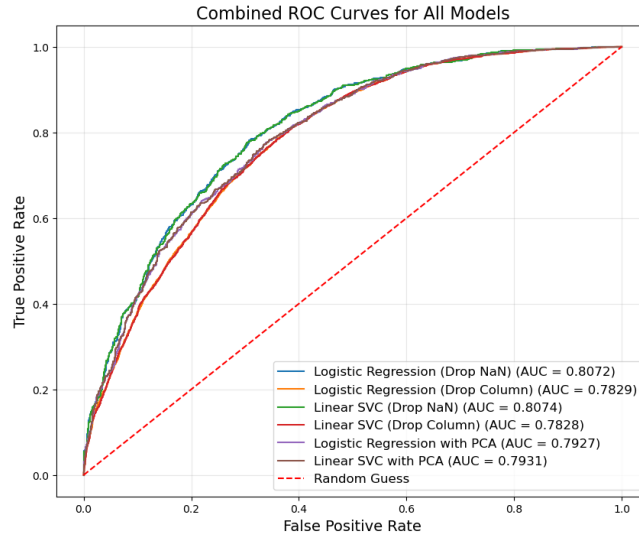


Figure 10: ROC-AUC for All Models

From a broader perspective, these findings highlight the complexity of predicting "pepper" ratings. The "Take Again Proportion" column likely captures critical elements of student sentiment toward teaching quality, making it a key feature despite its missing data. Moreover, the consistently high AUC values across models indicate that the dataset's tags and numerical factors effectively capture patterns in how students evaluate professors.

Extra Credit: Clustering

The implementation began by addressing the qualitative features in the dataset, specifically “Major/Field” and “University.” To ensure meaningful analysis, majors with fewer than 100 entries were excluded, resulting in a filtered dataset with 75 unique majors. This step helped eliminate noise from underrepresented fields while focusing on patterns from more commonly reported academic disciplines. After encoding “Major/Field” and “University” into numerical values using `LabelEncoder` and scaling these features, K-Means clustering was applied to group the professors into five clusters. The number of professors varied significantly across clusters, with Cluster 2 being the largest (15,805 professors) and Cluster 1 the smallest (7,988 professors). This uneven distribution highlights variability in how majors and universities are represented in the dataset.

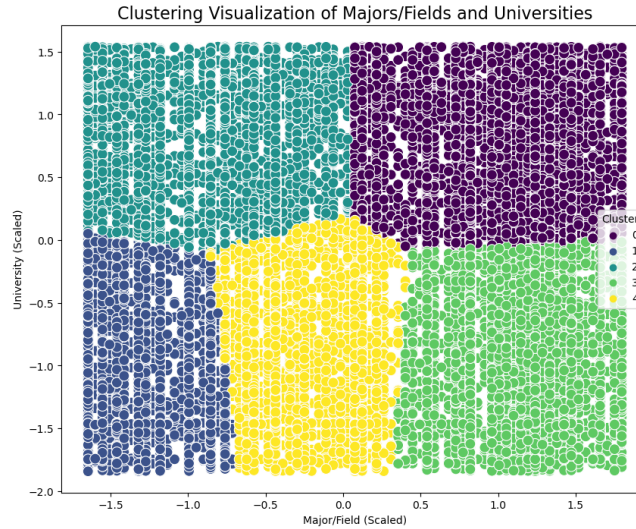
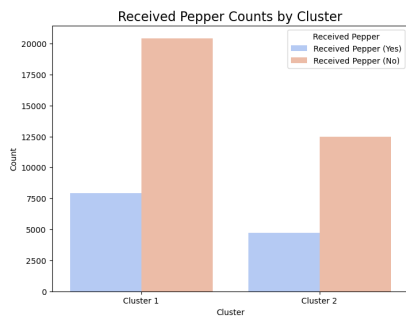


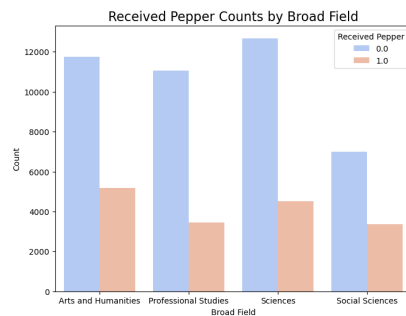
Figure 11: PCA of Major/Field and Universities

The visualization of the PCA-reduced data appeared as a grid because “Major/Field” and “University” are categorical features. After encoding and scaling, these features take on discrete numeric values, creating structured groupings. The grid-like appearance results from clustering patterns that reflect these discrete, encoded relationships. While this structure provides clarity in identifying clusters, it also underscores the limitations of applying PCA to categorical data, where the separation between points may be less meaningful than in continuous data.

Next, the relationship between cluster membership and whether professors received a pepper was tested using a chi-squared test. While the clustering revealed distinct groupings, the test showed no statistically significant association ($p = 0.0178$) between clusters and receiving a pepper. This indicates that clustering on “Major/Field” and “University” alone may not capture meaningful differences in how professors are evaluated by students.



(a) Received Pepper Counts by Cluster



(b) Received Pepper Counts by Broad Field

To refine the analysis, majors were grouped into broader academic categories: “Arts and Humanities,” “Professional Studies,” “Sciences,” and “Social Sciences.” This grouping revealed significant patterns, with a chi-squared test indicating a strong association ($\chi^2 = 315.48, p < 0.0001$) between academic field and the likelihood of receiving a pepper. Professors in “Arts and Humanities” and “Sciences” were most likely to receive peppers, suggesting that disciplines often associated with creativity, engagement, or scientific rigor may positively influence student evaluations. In contrast, “Professional Studies” and “Social Sciences” had comparatively lower pepper counts.

Appendix

Feature Correlation Matrix for Question 7

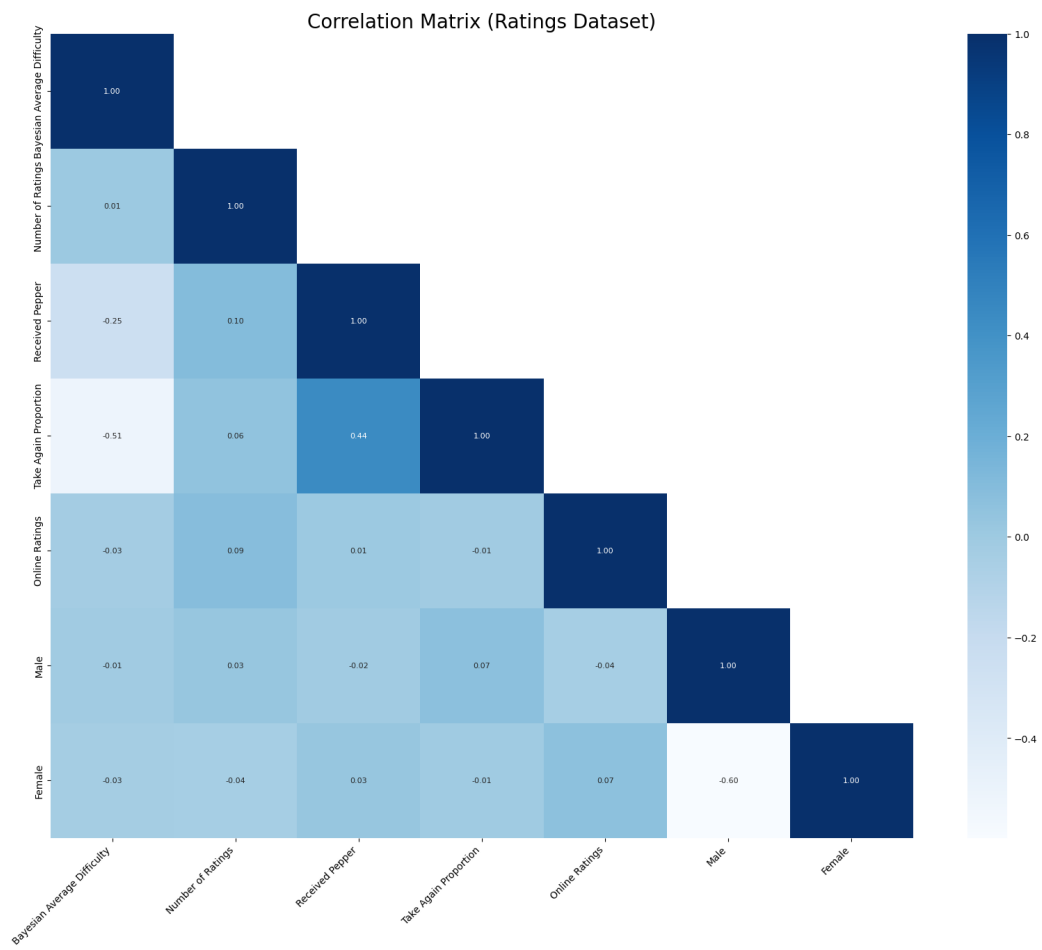


Figure 13: Correlation Matrix of Features for Question 7 (Ratings Dataset)